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A culture-derived rule for where spontaneous bursts ignite predicts an activity hub in the fly antennal lobe

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Neural networks are never silent. Dissociated cortical cultures generate spontaneous network bursts that start at preferred sites[1–3], but whether the wiring of an intact brain contains comparable ignition sites is unknown. In public high-density multielectrode recordings of seven developing rat cortical cultures (114 sessions, 30,349 bursts)[4], bursts began at hot spots that recurred from day to day; the leading eigenvector of between-burst functional coupling, not firing rate, predicted where they began, and activity stayed just below runaway (branching ratio ≈ 0.99). We then gave a whole-brain spiking model of the *Drosophila* connectome[5–7] the same conditions: weak noise and short-term synaptic depression. The behaviourally calibrated model did not burst; bursts appeared at 1.75–2.5-fold synaptic gain. Under three transmitter assumptions, the leading mode of the excitatory wiring lay in cholinergic local neurons of the antennal lobe. One type of them (lLN1_bc, 15 per side) was among the 0.15% most spontaneously active neurons and fired 30–45 ms before bursts or drove the brain tonically, and silencing the 111 cholinergic local neurons cut bursting 2.3–9-fold, whereas size-matched random sets did not. In resting-state imaging of 20 flies[8], the antennal lobe showed the slowest spontaneous fluctuations. We predict that these neurons are a hub of the fly's spontaneous activity.

Main

Spontaneous activity is a basic property of neural tissue. Its simplest form is seen in dissociated cortical neurons grown on multielectrode arrays (MEAs): within one to two weeks they self-organize into networks that fire brief, synchronized network bursts separated by seconds of sparse activity[1,9]. These bursts do not start at random. A small set of neurons or locations — leaders, or nucleation sites — fires first in most bursts and keeps doing so for hours[2,10–12]. Theory explains this as noise focusing: network topology funnels independent background noise into a few places where recurrent excitation converges, and those places ignite the whole network[3]. Cultured networks have also become a reference for collective dynamics near a critical point[9,13–15] and for sleep-like states that arise without a brain[16].

Whether these culture-level properties carry over to an intact brain is an open question. The adult *Drosophila* brain is now mapped synapse by synapse[5,6], and a leaky integrate-and-fire (LIF) model built from this wiring with a single free parameter predicts sensorimotor transformations that were confirmed experimentally[7]. Spontaneous activity has been approached from the connectome in three ways: structural connectivity predicts region-to-region correlations of resting-state calcium signals[8,17]; a rate model fitted to such data identified inhibitory hubs that keep spontaneous activity

bounded[18]; and eigen-decomposition of a linearized connectome model describes its intrinsic dynamical modes[19]. An earlier whole-brain model built from light-microscopy data showed that short-term synaptic depression stabilizes the network[20]. Where spontaneous population events start in the fly brain, and which neurons drive spontaneous activity, has not been addressed.

Here we asked the same three questions of both systems with the same measures: where do spontaneous bursts ignite, what predicts the ignition site, and how close to runaway does the network operate? We answered them first for cultured cortex using public high-density MEA recordings, which gave a rule — ignition follows the leading mode of the network's coupling rather than its firing rate. Applied to the fly connectome under culture-like conditions, the rule and direct simulation point to the same small population: cholinergic local neurons of the antennal lobe (AL). We then tested this prediction by silencing in the model and checked it against resting-state imaging of real flies.

Results

Bursts in cultured cortex start at recurring hot spots

We analysed DANDI dataset 002020 (ref. 4; Nakamuta et al.[21]): rat cortical cultures recorded on MaxOne high-density MEAs[22] for 30 min on most days of their development, with about 1,000 of 26,400 electrodes routed per session (Fig. 1a). After excluding 15 nearly empty sessions, 114 sessions from seven cultures contained 30,349 network bursts (median 1.6 bursts per min, interquartile range (IQR) 0.23–8.2; median duration 390 ms; median participation 94% of active electrodes; Methods). All parameters were fixed before cross-session results were inspected, and every headline number was checked at half and double the burst threshold (Methods).

Bursts began locally. The first 20 spikes of a burst covered 0.62 times the radius of 20 random spikes from the same burst (median ratio; below 1 in 69 of 70 sessions with at least 20 analysable bursts; $P = 4 \times 10^{-13}$), and the 10% of electrodes that fired first most often supplied the first spike of 81% of bursts, against 48% expected from where burst spikes occurred. The ignition points formed a few hot spots (Fig. 1b,c) that recurred within a culture: ignition maps of consecutive sessions correlated with $r = 0.43$, against $r = -0.06$ for pairs of sessions from different cultures (Mann–Whitney $P = 9 \times 10^{-18}$; higher in 7 of 7 cultures; Fig. 1d), and consecutive hot spots lay 0.58 mm apart against 1.75 mm between cultures. The ignition pattern was not a copy of where burst activity was densest; the residual map (ignition minus burst-activity density) was also stable ($r = 0.39$ versus 0.02, 7 of 7 cultures). Two caveats temper this result. Ignition points were only modestly more concentrated than centroids of random 20-spike windows from the same bursts (concentration 0.56 versus 0.48; 5 of 7 cultures), and consecutive sessions shared more of their electrode routing than sessions from different cultures (Jaccard overlap of covered area 0.49 versus 0.10).

Ignition follows the leading coupling mode, not firing rate

The noise-focusing idea predicts that bursts start where background activity converges. The simplest proxy, the firing rate of a location between bursts, did not predict ignition (Spearman ρ =

0.01 per 100-µm bin, median across 70 sessions; Fig. 1e). An apparent effect of local electrode density disappeared once we controlled for how many electrodes were routed nearby. By contrast, the weight of each location in the leading eigenvector of the between-burst functional-coupling matrix — the pattern along which coordinated background fluctuations are largest — predicted where bursts would start (ρ = 0.25, IQR 0.03–0.35; $P = 6 \times 10^{-8}$; positive in 7 of 7 cultures), beat the firing rate in 88% of sessions (56 of 64) and survived the routing control (partial ρ = 0.14, 7 of 7 cultures; Fig. 1e). Thus, in cultured cortex, bursts ignite where the network's coupling concentrates coordinated noise, not where neurons are merely most active. The effect is moderate, and on coarse 59-electrode arrays (200-µm pitch) it could not be resolved (Extended Data Fig. 1).

Cultured networks hover just below runaway

To place every system on one scale we estimated the branching ratio m — the average number of spikes triggered by one spike — with the subsampling-invariant multistep-regression estimator[15] on population activity in 4-ms bins. Cultured networks were close to, but below, runaway: m = 0.992 (IQR 0.985–0.997) in the HD-MEA cultures and 0.986 in the replication set (Wagenaar et al.[1]; 30 cultures, 526 recordings), with intrinsic timescales of a few hundred milliseconds. Removing the bursts (±100 ms) did not lower m (0.995 and 0.991), so the between-burst activity itself reverberates. With development, bursts became more frequent (7 of 7 and 29 of 30 cultures) and absorbed a growing share of spikes, while m moved away from 1 (whole session: 7 of 7 and 30 of 30 cultures; Fig. 1f and Extended Data Fig. 1). Within a 30-min session, m did not rise: it stayed flat in the HD-MEA cultures and fell very slightly in the replication set ($\Delta m = -0.0018$ between the first and second 15 min; $P = 8 \times 10^{-8}$), a fatigue-like rather than a potentiation-like drift[23].

The fly brain under culture-like conditions

We next gave the whole-brain LIF model of the FlyWire connectome (138,639 neurons, 15.1 million connections, 54.5 million synapses)[5–7] the two ingredients that make cultures burst: background noise and short-term synaptic depression. Every neuron received independent Poisson 'spontaneous spikes' at 0.3 Hz, and every synapse depressed with use (Tsodyks–Markram model[24]; utilization 0.05, recovery 2 s; Methods). A gain factor g multiplied all synaptic weights, with g = 1 being the published model whose single free parameter was calibrated against behaviour[7]. Because synaptic signs come from machine-learning transmitter predictions that are partly wrong[25], we ran every experiment under three sign assumptions (Methods): a minimal correction of the published signs (40 AL local neurons predicted with low confidence to be serotonergic, which act as spurious excitatory amplifiers, made inhibitory), one transmitter per cell type with literature-known transmitters taking precedence ('cell-type vote'), and the same without fast synaptic effects of monoamines ('no fast monoamines'). Without the minimal correction, any background noise drove the olfactory system into a self-sustained runaway state (Extended Data Fig. 2), so the published signs were not used further.

At the calibrated gain the model did not burst under the minimal correction or without fast monoamines, and produced only sparse events under the cell-type vote (Fig. 2a). Network bursts

appeared when the gain was raised to 1.75–2.0 (minimal correction), 2.0–2.5 (no fast monoamines) or 2.0 (cell-type vote; Fig. 2a), and became frequent at 2.5 (78–236 per min). Model bursts were brief (median 18–40 ms) and recruited a smaller share of neurons than culture bursts (Fig. 2b), and at g = 2.5 the model still emitted 79–95% of its spikes outside bursts, against a median of 53% in HD-MEA cultures (Extended Data Fig. 4). The calibrated fly brain therefore sits well inside the non-bursting regime, about a factor of two in synaptic gain away from culture-like bursting.

Bursts ignite in cholinergic antennal-lobe local neurons

To locate ignition we recorded every spike of the whole brain and asked, for each cell type in each hemisphere, whether it fired in excess during the 20 ms before burst onset and when it first crossed its own noise threshold (Methods). Under the minimal correction (29 bursts at g = 2.5), the earliest and most consistent groups were the cholinergic AL local neurons of type lLN1_bc in both hemispheres, which crossed threshold a median of 44 ms (left; 90% of bursts) and 30 ms (right; 55% of bursts) before brain-wide onset and fired three times more than expected in the pre-onset window (Fig. 2c,d). They were joined by lLN2X12, a second cholinergic-predicted AL local-neuron type (29 ms before onset). On a frontal map of the brain, neurons firing in excess before onset clustered in the two antennal lobes (Fig. 2c). The ignition site was largely preserved when the noise level was halved or doubled, depression was made stronger or other random seeds were used: across these six runs, a cholinergic AL local-neuron type was the earliest group in three and among the four earliest in five, and lLN1_bc crossed threshold 3–44 ms before onset in at least one hemisphere in all six (Extended Data Fig. 3).

Under the other two sign assumptions the AL cholinergic local neurons did not trigger bursts; instead they fired continuously at 60–136 Hz (lLN1_bc at g = 2.0–2.5), and bursts started in the visual system and the central complex (Extended Data Fig. 2). In all three assumptions, therefore, these neurons were among the first to become self-excited; what differed was whether they were still recovering between bursts (minimal correction) or already locked in a tonic state (cell-type vote, no fast monoamines). Consistent with this, lLN1_bc were among the most spontaneously active neurons of the whole brain even at the calibrated gain (3.2, 31 and 24 Hz under the three assumptions, against a median of 0.2 Hz; median rank within the top 0.15% of 138,639 neurons; Fig. 3c).

The leading mode of the fly's excitatory wiring

Why the antennal lobe? Following the culture result, we asked where the leading mode of the network's coupling lies — here computed directly from the wiring rather than from activity. The Perron eigenvector of the excitatory connectivity matrix, the pattern that grows first when excitation is scaled up, placed 28–34% of its weight on AL local neurons (8–13% on the 30 lLN1_bc neurons alone), 15–19% on AL projection neurons and 18–19% on Kenyon cells of the mushroom body under all three assumptions (Fig. 3a). With signed (excitatory and inhibitory) connectivity, the leading real mode under the cell-type vote and no-fast-monoamine assumptions was again centred on AL local neurons (37–39% of its weight). The 30 lLN1_bc neurons make about 112,000 output

synapses, mostly onto other AL local neurons and projection neurons, and 74–92% of their input synapses come from excitatory partners (depending on the transmitter assumption), so they form a dense, weakly inhibited recurrent excitatory loop. In the absence of short-term depression, they belonged to the first sizeable group of neurons that sustained activity on its own as the gain was raised: under every assumption, the first self-sustained core larger than eight neurons contained lLN1_bc neurons (5–30 of 30), with AL local neurons making up 14–29% of it alongside projection neurons, lateral-horn neurons and Kenyon cells (Fig. 3b and Extended Data Fig. 2).

Across all cell-type groups, however, the structural mode predicted ignition only about as well as firing rate did (Spearman ρ of Perron weight per neuron 0.12–0.32 versus 0.11–0.29 for the between-burst rate in six runs under the minimal correction; 0.25 versus 0.34 under the cell-type vote). The culture method itself — the leading eigenvector of simulated between-burst correlations — predicted ignition worse than either (ρ = −0.07 to 0.16; Fig. 4c and Extended Data Fig. 4), because its leading mode picked out strongly correlated, tonically active central-complex neurons. In the fly brain the rule therefore identifies the hub through the wiring, but does not rank the rest of the brain better than firing rate does.

Silencing the hub suppresses spontaneous bursting

If the AL cholinergic local neurons are the brain's ignition hub, removing them should suppress spontaneous bursting more than removing any comparable set of neurons. We silenced the outputs of the 111 AL local neurons with cholinergic cell-type identity, of the 30 lLN1_bc neurons alone, of all 5,177 Kenyon cells, of size-matched random neurons and of size-matched AL local neurons of other transmitters (Fig. 3d). Silencing the 111 cholinergic AL local neurons reduced spontaneous bursting 2.3–9.0-fold under all three assumptions and at both gains tested (relative burst rate 0.11–0.43), below every one of 15 size-matched control sets in each condition (0.74–1.74; Fig. 3d). Silencing only the 30 lLN1_bc neurons reduced bursting 1.6–3.2-fold under the minimal correction and without fast monoamines (0.32–0.63, again below all controls) but not under the cell-type vote (1.06–1.11), where the other cholinergic local neurons sufficed. Silencing all 5,177 Kenyon cells changed bursting by factors of only 0.75–1.12. Silencing the 30 lLN1_bc neurons — 0.02% of the brain — also lowered the brain-wide mean firing rate by 11–21%, whereas silencing 5,177 random neurons lowered it by 1–4% (Fig. 3e). The hub did not depend on the disputed chemical synapses from these neurons onto projection neurons (see Discussion): after removing all 90,491 such synapses, the cholinergic local neurons still carried 21% (minimal correction) and 56% (no fast monoamines) of the Perron weight, they made up 34% and 91% of the first self-sustained core, and silencing lLN1_bc still reduced bursting 2.4-fold without fast monoamines (38.8 to 16.3 per min; 30 random neurons, 36.7), although only by 12% under the minimal correction (Extended Data Fig. 3).

The antennal lobe has the slowest spontaneous fluctuations in real flies

The model predicts that the antennal lobe should stand out in spontaneous activity. We tested this in resting-state two-photon imaging of 20 flies expressing pan-neuronal GCaMP6s without stimulation (Turner et al.[8]; 1.2 volumes per s, about 28 min per fly, 74 atlas regions). Of all regions,

the right antennal lobe had the highest lag-1 autocorrelation and the left the fourth highest (Fig. 4a). Across flies, the antennal lobes fluctuated more slowly (lag-1 autocorrelation 0.85 versus a median of 0.58 for other regions; higher in 18 of 20 flies; Wilcoxon $P = 1 \times 10^{-5}$) and with larger amplitude (s.d. of $\Delta F/F$ 0.048 versus 0.029; 19 of 20 flies; $P = 1 \times 10^{-5}$) than the rest of the brain, while being slightly less coupled to the brain-wide signal (correlation with the mean of other regions 0.50 versus 0.52; 5 of 20 flies higher; $P = 0.04$; Fig. 4b). Left and right antennal lobes were only weakly correlated with each other after removing the global signal (partial $r = 0.15$), so each lobe fluctuates largely on its own. This is what an autonomous generator in each lobe would produce, although slow sensory input to the antennae could contribute as well. A region-by-region comparison between model burst participation and real fluctuation measures was weak ($|\rho| \leq 0.30$ across 73 regions; Extended Data Fig. 4).

Bursts do not preferentially reach the sleep circuits

Because synchronized bursting in cultures has been likened to sleep[16], we asked whether model bursts preferentially recruit the fly's sleep-regulating neurons: the R5 ring neurons, which show sleep-need-dependent slow-wave synchrony[26], the dorsal fan-shaped body[27,28], helicon cells[27] and clock neurons. Relative to random central-brain cell types, none of these was consistently over-recruited across sign assumptions (Extended Data Fig. 4). The simulated bursts thus do not provide a direct route from spontaneous ignition to the sleep homeostat.

Discussion

Cultured networks and the fly connectome gave a consistent answer to where spontaneous bursts begin: at the leading mode of the network's coupling rather than at its most active elements. In cultured cortex the mode was read from coordinated background fluctuations and predicted ignition hot spots that recurred over days. In the fly the mode was read directly from the wiring diagram, and it located a compact recurrent loop of cholinergic AL local neurons that was among the first parts of the brain to self-excite, that led or tonically drove spontaneous bursts, and whose silencing suppressed bursting far more than silencing size-matched or much larger sets of other neurons. That the same small population came out of structure, direct simulation and silencing, under three transmitter assumptions and in both hemispheres, is the main reason to take the prediction seriously.

The prediction is consistent with what is known about these cells. Cholinergic local neurons were first described as about 15 cells per antennal lobe that spread diffuse excitation across glomeruli[29,30], and paired recordings found reciprocal excitatory connections between them and with projection neurons[31]. Two findings constrain the interpretation. First, their excitation of projection neurons is largely electrical[32], whereas the connectome contains only chemical synapses; our control shows, however, that the hub persists structurally without these synapses, carried by connections among the local neurons themselves, although its causal weight then depends on the transmitter assumption. Second, in vivo these neurons are barraged by spontaneous inhibition[32], interglomerular inhibition outweighs excitation[33], and at rest projection-neuron activity is driven mainly by receptor input rather than by network activity[34]. Our model

agrees that the calibrated network does not burst; the prediction concerns what happens when excitability rises or inhibition weakens. Notably, the first olfactory cortex is the most seizure-prone region of the mammalian brain[35–37], suggesting that dense recurrent excitation in early olfactory circuits may make them the weakest point for runaway activity in both lineages.

The results suggest four experiments. (1) Whole-cell or voltage-imaging recordings should find lLN1_bc-type neurons among the most spontaneously active central neurons; their spontaneous rates in vivo have, to our knowledge, not been reported. (2) After pharmacological disinhibition of the antennal lobe (picrotoxin with CGP54626[33,38]) or other manipulations that raise excitability, spontaneous synchronous events should appear first in the antennal lobe and should be preceded by activity of these neurons. (3) Silencing them genetically should reduce spontaneous fluctuations in the antennal lobe, lateral horn and mushroom body in resting-state imaging[8,17]. (4) In cultures, the leading coupling mode computed from a few minutes of between-burst activity predicts tomorrow's ignition hot spots, which can be tested by patterned optogenetic stimulation of predicted versus control sites.

The culture results also bear on how close to runaway a network operates and how this changes with use. Cultured networks sat at $m \approx 0.99$, consistent with previous estimates[15], but developed away from $m = 1$ as bursts took over, in line with the idea that strong bursting and reverberating background activity trade off[39]. Within a session we saw no build-up of m, which argues against a synaptic-homeostasis-like increase of effective coupling on a 30-min timescale[23], although a fatigue-like decrease was tiny. The fly model, by contrast, needed about twice its calibrated synaptic gain to burst; wake-associated synaptic growth in flies[40] is far smaller, so if it raises the brain's propensity for spontaneous bursts, the antennal lobe is where that should show first.

This study has clear limitations. The LIF model has no gap junctions, neuromodulation, adaptation or graded transmission; noise and depression were uniform; and transmitter identity is uncertain, which is why ignition differed between assumptions in whether the hub triggered bursts or fired tonically. The Perron mode was concentrated on the left antennal lobe (80% of its weight), although ignition and silencing effects occurred on both sides; left–right differences in this region may partly reflect reconstruction asymmetry. Culture data were unsorted electrode-level detections with unknown days in vitro, and ignition maps were partly confounded by electrode routing. The fly imaging data have a temporal resolution of about 0.8 s, far too slow to resolve ignition directly. Despite these limits, the convergence of culture data, connectome structure, simulation and resting-state imaging on one small population of neurons makes the cholinergic local neurons of the antennal lobe a concrete, testable candidate hub of the fly's spontaneous activity.

Methods

Cultured-network data

HD-MEA dataset. DANDI 002020 (version 0.261004.1410)[4,21] contains spontaneous activity of seven dissociated rat (E18) cortical cultures recorded on MaxOne HD-MEAs (26,400 electrodes, 17.5-µm pitch, about 1,000 routed per session) for 30 min on 11–30 days per culture (129

sessions). Spikes are channel-level threshold detections (not spike-sorted) with electrode coordinates. Days in vitro are not provided; we used days since each culture's first recording, so only within-culture trends were interpreted. Channels with at least 0.1 Hz were active. Sessions with fewer than 30,000 spikes (scaled to channel number and duration) were excluded (15 sessions).

Replication dataset. Spike times of 30 dense rat cortical cultures recorded on 59-electrode MEAs (200-µm pitch) at 3–39 days in vitro (527 recordings, about 45 min each) were obtained from Wagenaar et al.[1]; channel numbers were converted to grid positions with the authors' tables.

Network bursts. Population spike counts in 10-ms bins; a burst was a maximal run of bins at or above half of a high threshold, max(5 × median count, 0.1 × number of active channels, 5), containing at least one bin at or above that threshold; runs separated by less than 100 ms were merged, and events recruiting less than 20% of active channels were not counted as bursts. This recruitment rule was added after inspecting four traces and before any cross-session analysis. All parameters were fixed across sessions; sensitivity was tested by halving and doubling each.

Ignition. The ignition point of a burst was the centroid of its first 20 spikes. Ignition maps were counts on a 100-µm grid (Gaussian smoothing σ = 100 µm) normalized by the number of bursts; analyses required at least 20 bursts per session. Stability was the Pearson correlation of maps of consecutive analysable sessions over grid bins covered in both, compared with all pairs of sessions from different cultures. Predictors were computed per bin and correlated with ignition probability by Spearman's ρ: between-burst firing rate (bins farther than 100 ms from any burst), in-burst rate, local density and the leading eigenvector of between-burst functional coupling. For the last, between-burst spike counts in 20-ms bins were correlated (Pearson) between channels with at least 30 between-burst spikes; negative values, the diagonal and pairs closer than 50 µm (possible duplicate detections) were set to zero, and the leading eigenvector was taken. Partial correlations controlled for the number of routed active electrodes within 200 µm.

Branching ratio. The branching ratio m was estimated by multistep regression[15]: for population counts A_t in 4-ms bins, slopes r_k of A_{t+k} on A_t for k = 1–50 were fitted with $r_k = b\,m^k$; $\tau = -4$ ms/ln m. The between-burst estimate used segments of at least 1 s outside bursts ±100 ms. The same estimator (one implementation) was used for cultures and model.

Whole-brain model

Model and simulator. We used the LIF model of Shiu et al.[7,41] with the FlyWire v783 connectome[5,6] and its published constants (resting and reset potential −52 mV, threshold −45 mV, membrane time constant 20 ms, synaptic time constant 5 ms, refractory period 2.2 ms, delay 1.8 ms, 0.275 mV per synapse). Our event-driven GPU implementation reproduces the original Brian 2 code[42] neuron by neuron (Pearson $r \geq 0.999$), including the discarding of synaptic input during the refractory period. It was extended with: (1) spontaneous drive, independent Poisson suprathreshold kicks to every neuron (0.3 Hz unless stated); (2) a gain g multiplying all synaptic weights; (3) Tsodyks–Markram short-term depression of each neuron's outputs[24] (each spike

transmits with the current resource x, which drops by a fraction $U = 0.05$ and recovers with $\tau = 2$ s); and (4) per-trial silencing, which removes a neuron's outgoing synaptic events. Time step 0.1 ms.

Transmitter assumptions. Synaptic signs follow transmitter identity (acetylcholine excitatory; GABA, glutamate and histamine inhibitory; monoamines excitatory as in ref. 7). 'Minimal correction': the published signs, except that 40 AL local neurons predicted with low confidence to be serotonergic[25] were made inhibitory. 'Cell-type vote': one transmitter per cell type (sum of prediction confidences over members), overridden by literature-known transmitters annotated in FlyWire[6] (also voted per cell type); 15,901 neurons changed sign relative to the published model. 'No fast monoamines': as the cell-type vote, but dopaminergic, serotonergic, octopaminergic and tyraminergic neurons had no fast synaptic effect.

Self-sustained activity. Without depression, all neurons received 0.5-Hz drive for 500 ms; neurons firing at least 10 Hz 1–2 s after the drive stopped formed the self-sustained core (three seeds per gain).

Bursts and ignition. Bursts were runs of at least 10 ms in which the 20-ms smoothed population rate exceeded 1.5 times the expected background (number of neurons × drive rate) plus 30 spikes per ms. For ignition, onsets were refined to the last millisecond at which the 5-ms smoothed rate was below 1.25 times background plus 10 spikes per ms. For every cell type × hemisphere group, pre-onset excess compared spikes in the 20 ms before onset, summed over bursts, with the number expected from the group's rate at least 200 ms from any burst (one-sided Poisson test); the crossing time was the first millisecond, from 50 ms before onset, at which the group's 5-ms spike count exceeded its baseline by five Poisson standard deviations and three spikes. Ignition runs were 10 s; burst statistics used 15-s runs with two seeds.

Structural modes. The leading eigenvector of the excitatory connectivity (synapse counts from excitatory neurons; Perron mode) and the eigenvalues with the largest real part of the signed connectivity were computed with ARPACK. Weights were summed per cell type × hemisphere and divided by group size for correlation with ignition.

Silencing. At gains around and above burst onset, outputs of a neuron set were removed and bursts were counted (8-s runs, two seeds, five gains; confirmation: 20-s runs, three seeds, two gains and five draws of each control). Sets: the 111 AL local neurons with cholinergic identity under the cell-type vote; the 30 lLN1_bc neurons; all 5,177 Kenyon cells; random neurons and AL local neurons of other transmitters matched in number. The eLN→PN control removed all chemical synapses from the cholinergic AL local neurons onto AL projection neurons (90,491 synapses).

Sleep circuits. In-burst peak over baseline activity was computed for ER5 ring neurons, dorsal fan-shaped body types FB6 and FB7, helicon cells (ExR1), DN1 and l-LNv clock neurons, PPL1 dopaminergic neurons and mushroom-body output neurons, and compared with 500 random draws of the same number of central-brain cell types.

Virtual MEA. Spikes of 1,024 randomly chosen neurons were analysed with the culture measures (out-of-burst fraction, branching ratio).

Resting-state imaging of flies

Region time series of Turner et al.[8] (20 flies, nSyb-GAL4 > GCaMP6s, about 1.2 Hz volumetric two-photon imaging, no stimulus, Ito atlas regions) were loaded with a restricted unpickler. Δ*F*/*F* used an 8th-percentile baseline in a 60-s window. Per region we computed the s.d., excess kurtosis, lag-1 autocorrelation, correlation with the mean of all other regions, and mean z-scored activity in frames where the global signal was in its top 5%. Antennal lobes were compared with the median of other regions within each fly (Wilcoxon signed-rank test). Model burst participation per neuropil used each neuron's fraction of synapses in each FlyWire neuropil.

Statistics

Culture statistics are medians across sessions with IQRs; because sessions of one culture are not independent, every claim is also reported as the number of cultures in which the effect had the same sign. Tests are two-sided unless stated.

Data and code availability

All data are public: DANDI 002020 (ref. 4), the Wagenaar et al. dataset[1], FlyWire v783 connectivity and annotations[5,6,41] and the Turner et al. imaging data[8]. All code, intermediate results and figure scripts are in the GitHub repository fly_culture_spontaneous.

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Author contributions

K.T. conceived the study; analyses and writing were carried out with Claude (Anthropic), an AI assistant.

Competing interests

The author declares no competing interests.

Figures

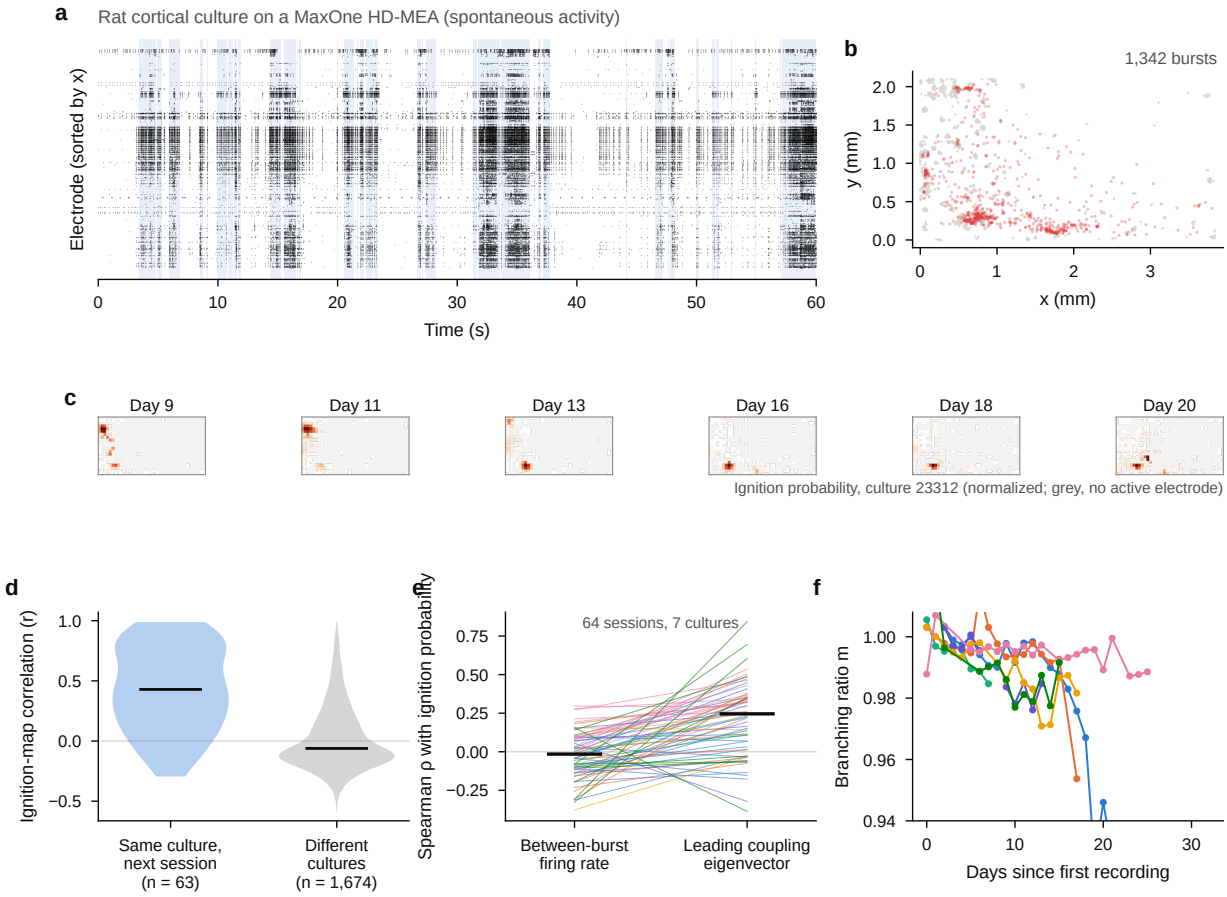


Fig. 1 | In cultured cortex, spontaneous bursts ignite at recurring hot spots predicted by the leading coupling mode. a, Spontaneous activity of a rat cortical culture on a MaxOne HD-MEA (culture 23312; 60 s; half of the active electrodes shown, sorted by x position); shading, detected network bursts. **b**, Ignition points (centroid of the first 20 spikes) of all 1,342 bursts of this session (red) over active electrodes (grey). **c**, Ignition-probability maps of the same culture on six days (normalized per map; grey, no active electrode). **d**, Correlation of ignition maps between consecutive sessions of the same culture and between sessions of different cultures; bars, medians. **e**, Spearman correlation between ignition probability and two predictors per 100-µm bin: between-burst firing rate and the leading eigenvector of between-burst functional coupling; lines, sessions (colour, culture); bars, medians. **f**, Branching ratio *m* (multistep regression, 4-ms bins) of each culture versus days since its first recording.

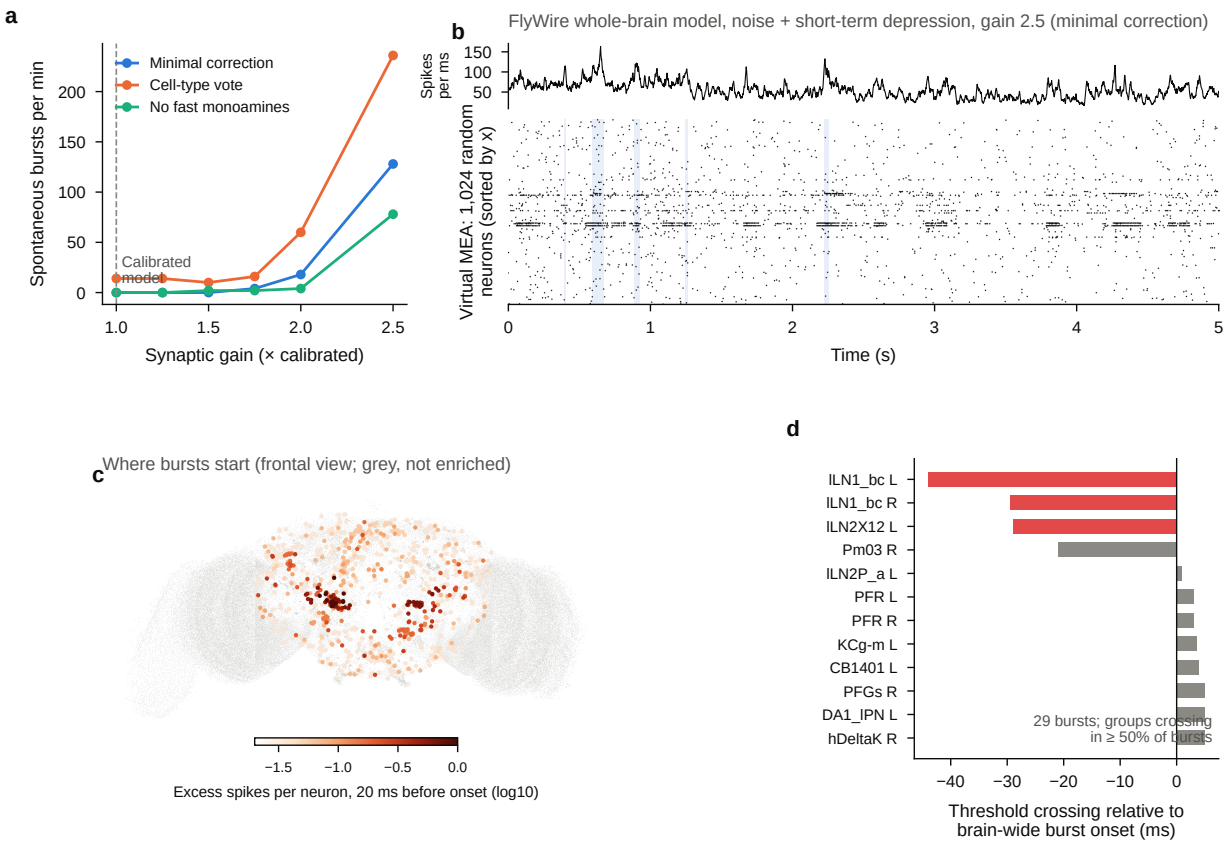


Fig. 2 | Under culture-like conditions the fly whole-brain model bursts, and bursts ignite in the antennal lobe. a, Spontaneous bursts per minute versus synaptic gain (1 = behaviourally calibrated model) under three transmitter assumptions (mean of two 15-s runs). **b**, Whole-brain population rate (top) and spikes of a 'virtual MEA' of 1,024 random neurons (bottom) at gain 2.5 under the minimal correction; shading, detected bursts. **c**, Frontal view of all neurons; coloured, neurons of cell-type groups that fired in excess in the 20 ms before burst onset (Poisson $P < 10^{-6}$), coloured by excess spikes per neuron; the dark clusters are cholinergic AL local neurons. **d**, Median time at which each group first crossed its noise threshold relative to brain-wide onset, for the 12 earliest groups that crossed in at least half of the 29 bursts; red, cholinergic AL local neurons.

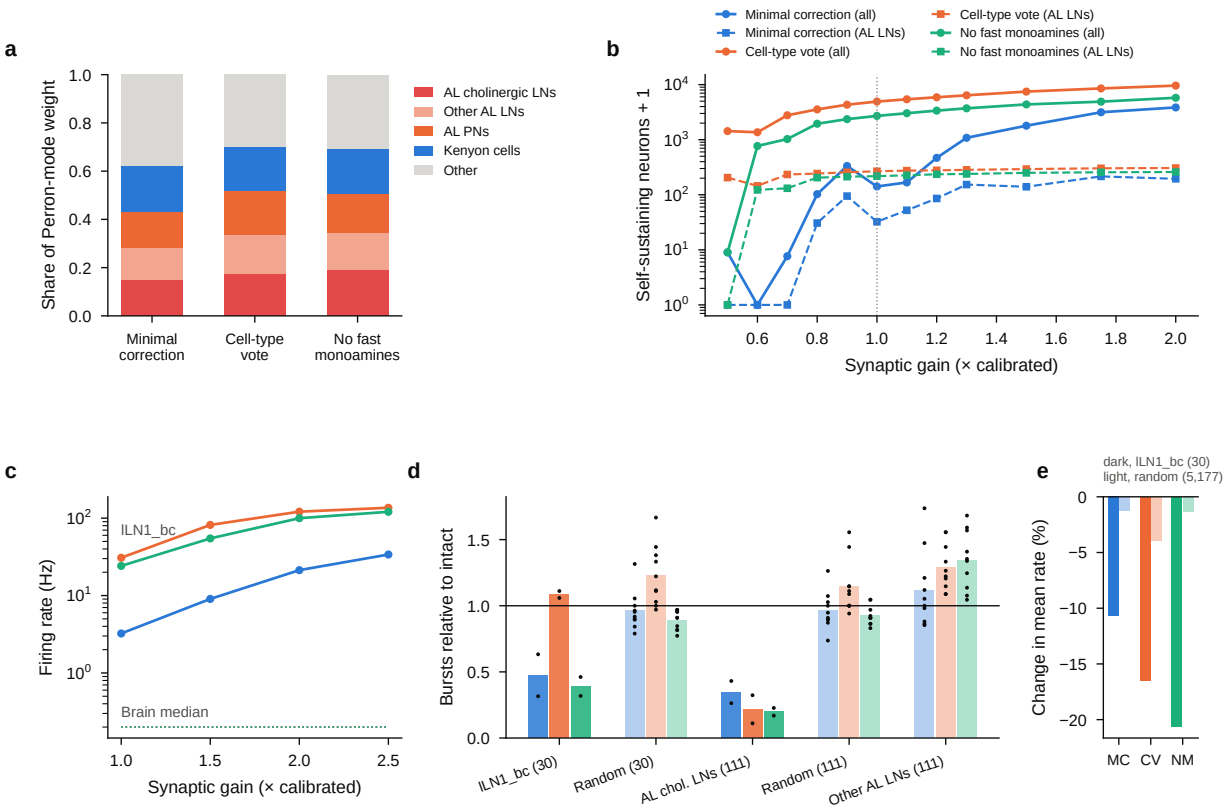


Fig. 3 | Cholinergic antennal-lobe local neurons form the leading mode of the excitatory wiring and a hub of spontaneous activity. a, Weight of the Perron eigenvector of the excitatory connectivity by cell class under three transmitter assumptions. **b**, Number of neurons that sustain activity after the drive stops (no short-term depression) versus synaptic gain; colour, AL local neurons, Kenyon cells and others (mean of three seeds). **c**, Spontaneous firing of lLN1_bc neurons, all cholinergic AL local neurons, AL projection neurons, Kenyon cells and the brain-wide median versus gain. **d**, Spontaneous bursts per minute after silencing the outputs of the indicated sets, relative to the intact model at the same gain (20-s runs, three seeds; points, the two gains per assumption, and for controls each of five random draws; bars, means). Colour, transmitter assumption. **e**, Change in brain-wide mean firing rate after silencing the 30 lLN1_bc neurons (dark) or 5,177 random neurons (light) at the highest gain tested (MC, minimal correction; CV, cell-type vote; NM, no fast monoamines).

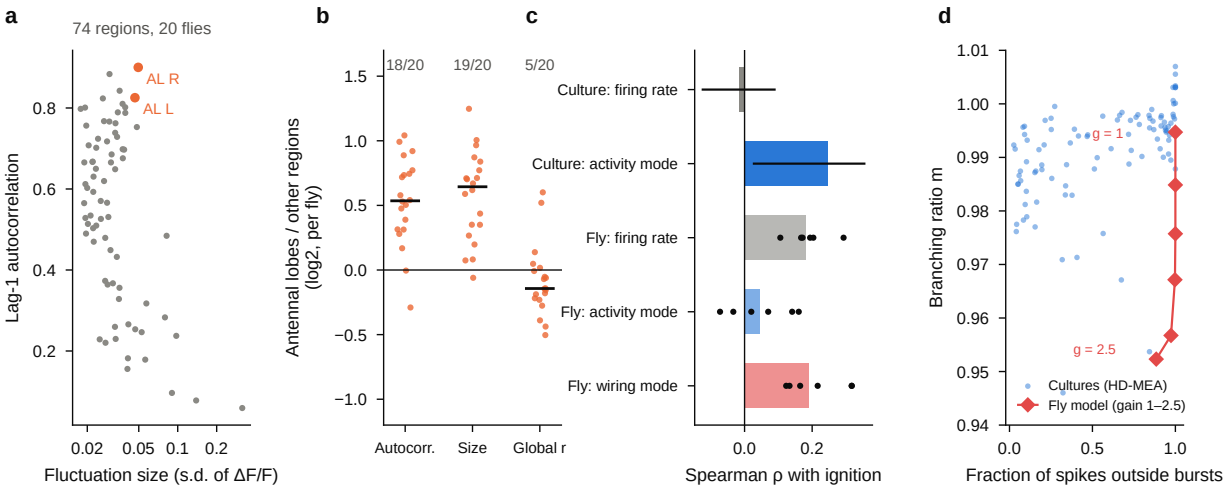
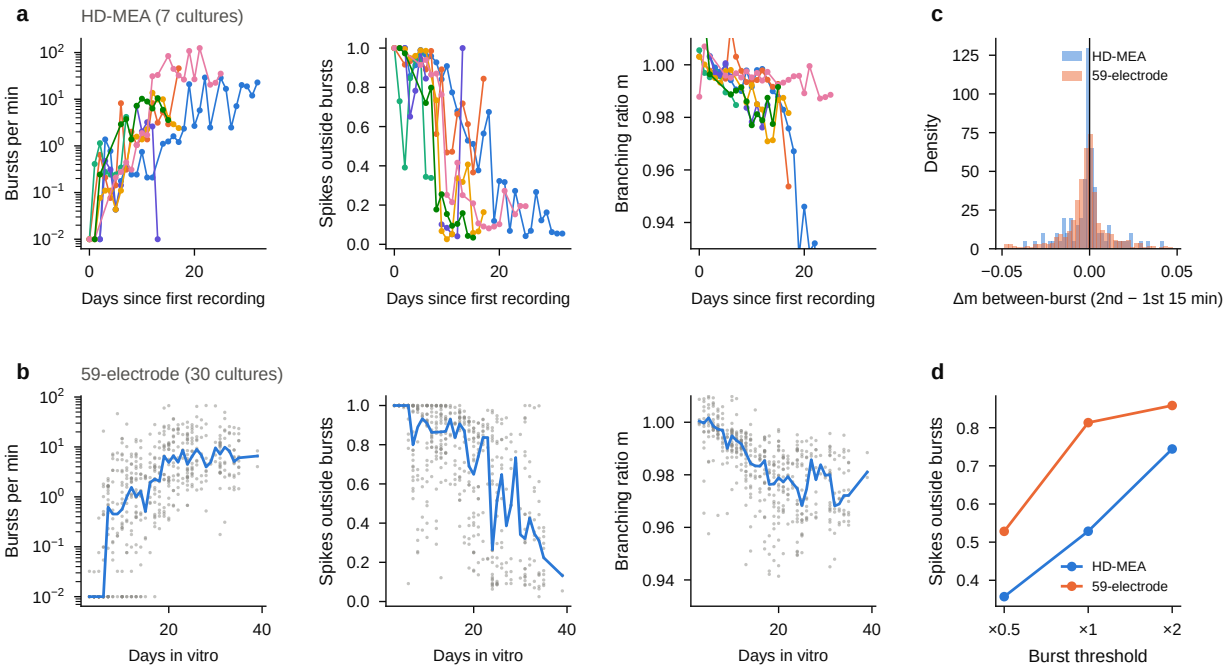
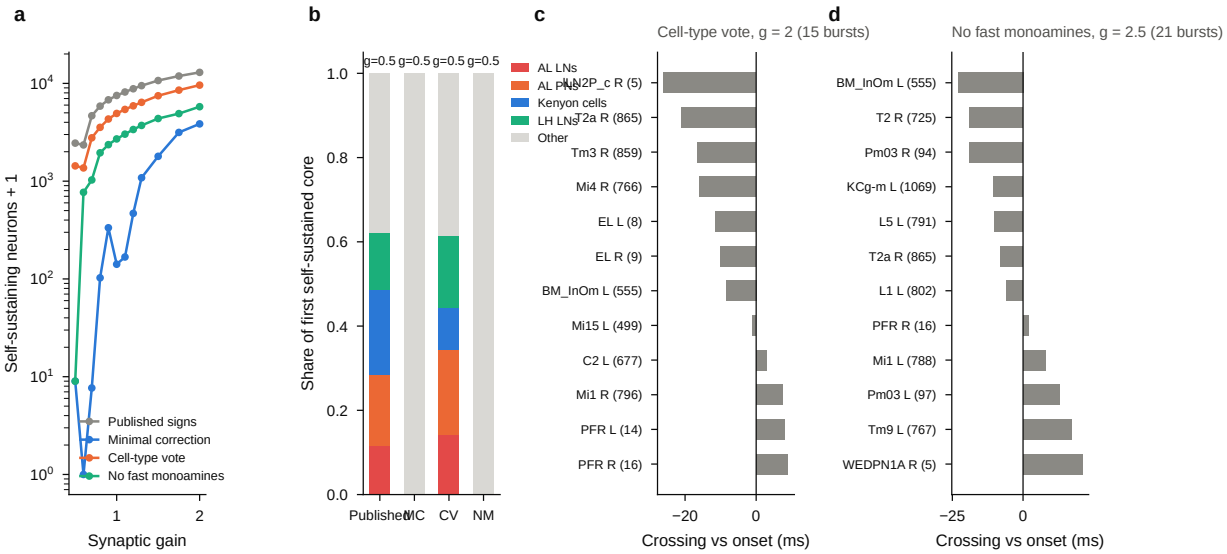


Fig. 4 | In real flies the antennal lobe shows the slowest spontaneous fluctuations, and a common scale for cultures and the fly model. a, Lag-1 autocorrelation versus s.d. of Δ*F*/*F* for 74 brain regions in resting-state imaging of 20 flies (median across flies; data from Turner et al.[8]); orange, antennal lobes. **b**, Antennal lobes versus the median of other regions in each fly for autocorrelation, fluctuation size and correlation with the global signal; lines, flies. **c**, What predicts ignition: Spearman ρ of between-burst firing rate and of the leading coupling mode with ignition, in cultures (per 100-µm bin; activity-based mode; bars, medians over 64 sessions) and in the fly model (per cell-type group; activity-based and wiring-based modes; dots, six runs under the minimal correction). **d**, Branching ratio versus the fraction of spikes outside bursts for culture sessions (dots) and for the fly model's virtual MEA across gains (diamonds; minimal correction).

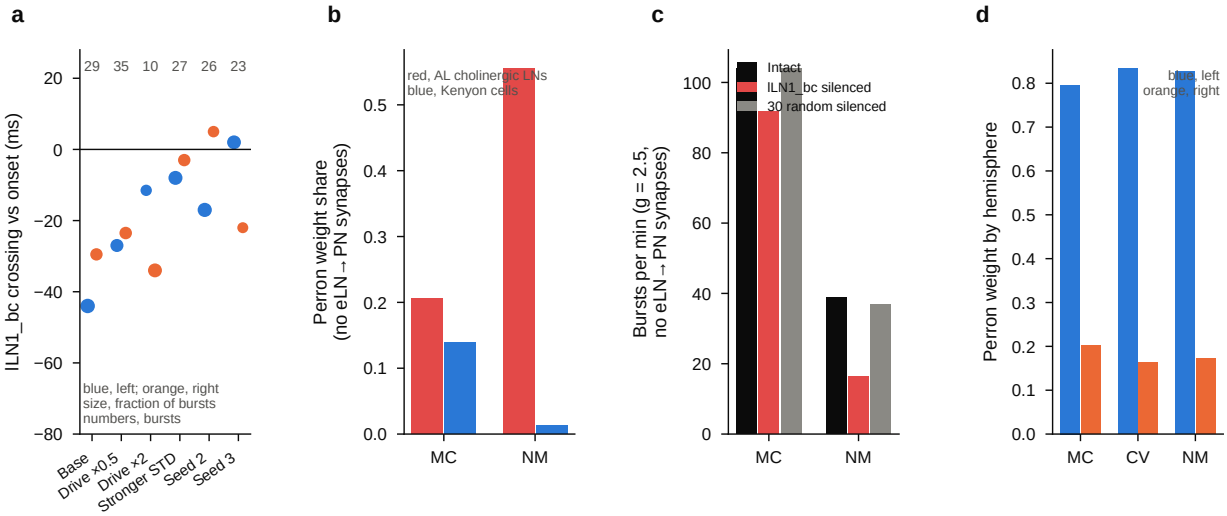
Extended Data



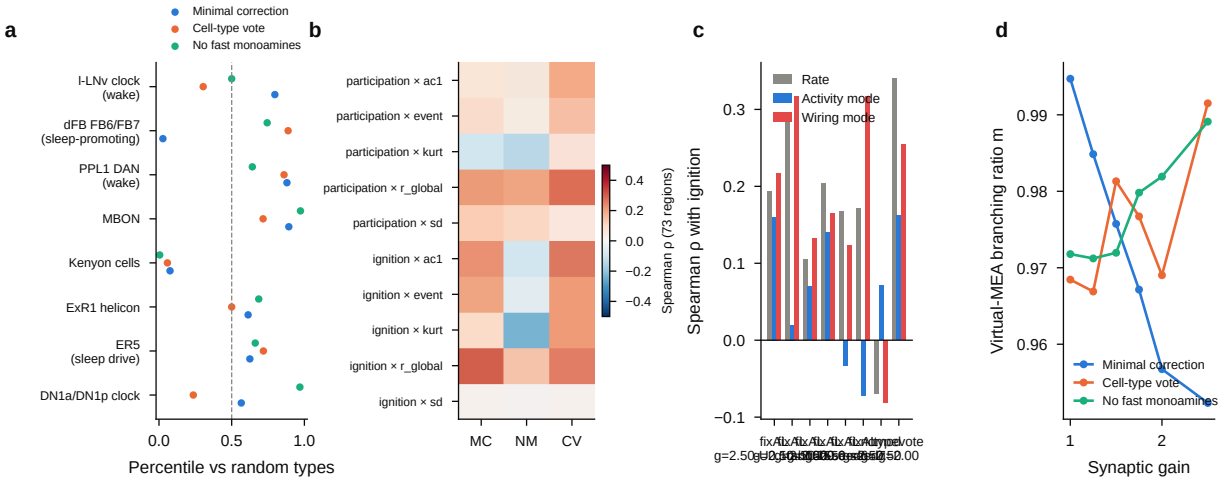
Extended Data Fig. 1 | Development, drift and replication in cultured networks. a,**b**, Burst rate and fraction of spikes outside bursts versus days since first recording (HD-MEA cultures) and days in vitro (59-electrode replication set). **c**, Branching ratio versus days in vitro in the replication set (medians per day). **d**, Within-session change of the between-burst branching ratio (second minus first 15 min) in both datasets. **e**, Sensitivity of burst rate and out-of-burst fraction to halving and doubling the burst threshold.



Extended Data Fig. 2 | Transmitter assumptions. **a**, Size of the self-sustained core versus gain under the published signs and the three assumptions used. **b**, Composition of the self-sustained core at the lowest gain at which it appears. **c**,**d**, Earliest-crossing groups in bursts under the cell-type vote (gain 2.0) and without fast monoamines (gain 2.5).



Extended Data Fig. 3 | Robustness of the ignition hub. **a**, Crossing time of lLN1_bc (left and right) relative to burst onset for different drive rates, depression parameters and seeds (minimal correction, gain 2.5). **b**, Perron weight by cell class after removing all chemical synapses from cholinergic AL local neurons onto projection neurons. **c**, Bursts and mean rate in that network with lLN1_bc or 30 random neurons silenced. **d**, Left–right distribution of the Perron weight.



Extended Data Fig. 4 | Weak and negative results. a, Recruitment of sleep and clock circuits by model bursts, as a percentile of random central-brain cell types of the same number (0.5, chance). **b**, Spearman correlation across 73 regions between model burst participation or pre-onset ignition and real resting-state measures. **c**, Activity-based leading mode versus between-burst rate in predicting ignition in the fly model. **d**, Virtual-MEA branching ratio and out-of-burst fraction of the model versus gain.